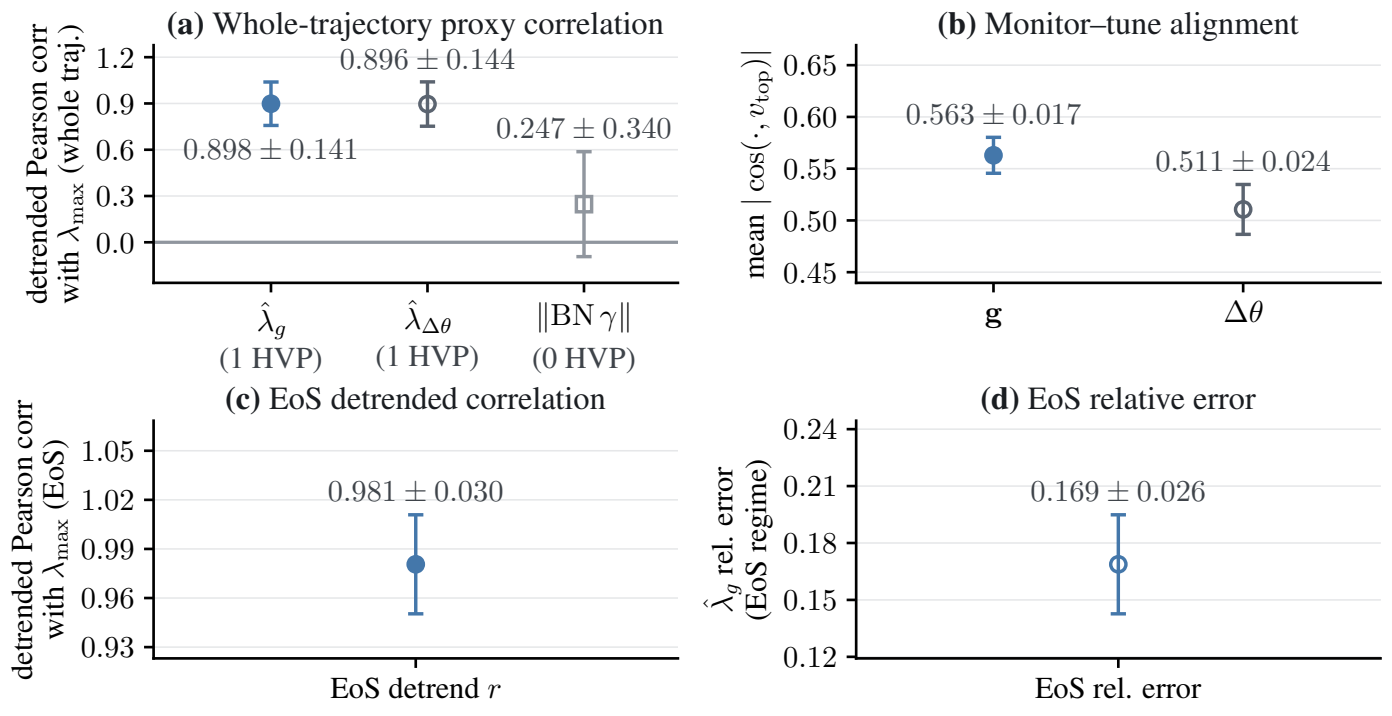

(a) Whole-trajectory proxy correlation
detrended Pearson corr with $\lambda_{\max}$ (whole traj.)
1.2
0.9
0.6
0.3
0.0
0.898 ± 0.141
0.896 ± 0.144
0.247 ± 0.340
$\hat{\lambda}_g$ (1 HVP)
$\hat{\lambda}_{\Delta\theta}$ (1 HVP)
$\|\mathrm{BN}\,\gamma\|$ (0 HVP)
(b) Monitor–tune alignment
mean $|\cos(\cdot, v_{\mathrm{top}})|$
0.65
0.60
0.55
0.50
0.45
0.563 ± 0.017
0.511 ± 0.024
g
$\Delta\theta$
(c) EoS detrended correlation
detrended Pearson corr with $\lambda_{\max}$ (EoS)
1.05
1.02
0.99
0.96
0.93
0.981 ± 0.030
EoS detrend r
(d) EoS relative error
$\hat{\lambda}_g$ rel. error (EoS regime)
0.24
0.21
0.18
0.15
0.12
0.169 ± 0.026
EoS rel. error